Instructions:

- Solutions are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

Problem 1. Bending Magnets 15 pts.

Suppose that we want to design a synchrotron to accelerate protons from an injection energy of 50 GeV to 300 GeV. The circumference of the ring will be $C = 4000$ m composed of 120 sections that typically contain 8 bending magnets and 2 quadrupole magnets. The bending magnets are left out of 20 sections to provide space for injection, extraction, acceleration, vacuum pumping, etc. Bending magnets in the remaining section each have an effective length of $\ell = 3.25$ m. Assume sector magnets.

a) 2 pts: what is the fraction of the ring circumfrance C filled by bending magnets?

b) 2 pts: What is the angular deflection per dipole in milliradians?

c) 3 pts: What is the "average bend radius" $\equiv C/(2\pi)$ of the ring? What is the actual bending radius of the dipole magnets? Why are they different?

d) 3 pts: Evaluate the needed range of magnetic fields in the bending magnets from injection to final energy.

e) 5 pts: Iron dipole bending magnets are employed. The magnets have 10 cm gaps between iron poles. How many amp-turns, $N \cdot I$, are required to drive the magnet at the maximum required field?

SOLUTION: Bending Magnets 15 pts.

a) 2 pts:

$$\text{fraction} = \frac{100 \text{ sectors} \cdot 8 \text{ magnets} \cdot 3.25 \text{ m}}{4000 \text{ m}} = 0.65$$

b) 2 pts:

$$\theta_{\text{bend}} = \frac{2\pi}{100 \text{ sectors} \cdot 8 \text{ magnets}} = 7.85 \text{ mrad}$$

c) 3 pts:

$$R_{\text{avg}} = \frac{\text{circum}}{2\pi} = 637 \text{ m}$$

$$\rho = \frac{\ell_{\text{magnet}}}{\theta_{\text{bend}}} = 414 \text{ m}$$

$\rho < R_{\text{avg}}$ because the filling factor of dipoles is less than 1.

d) 3 pts:

$$B_0 = \frac{[B\rho]}{\rho} \tag{1}$$

$$[B\rho] = \frac{p}{q} \tag{2}$$

$$E^2 = c^2p^2 + m^2c^4 \quad p = \sqrt{\frac{E^2}{c^2} - m^2c^2} \tag{3}$$

$$B_0 = \frac{p}{q\rho} = \frac{\sqrt{\frac{E^2}{c^2} - m^2c^2}}{q\rho} \tag{4}$$

Energy (GeV)	Magnetic Field (T)
50	0.403
300	2.419

e) 5 pts: Using the results from class:

$$B_0 = \frac{\mu_0(NI)}{g}$$

$$\Longrightarrow \quad (NI)_{\max} = \frac{0.1 \text{ m} \cdot 2.419 \text{ T}}{\mu_0} = 192 \text{ kA}$$

Problem 2. Muon Storage ring 20 points

Consider muon storage rings.

a) 5 pts: What is the total power radiated in an isomagnetic muon storage ring of radius ρ with an average beam current of I?

b) 5 pts: How much power is emitted in a 30 GeV ring with $\rho = 250$ m that stores 1 A of electrons? For 1 A of muons?

c) 5 pts: What is the natural beam current lifetime decay for 30 GeV muons? (Hint: muon decay time is roughly $\tau_\mu = 2.2$ μs.)

d) 5 pts: What is the heat load due to the decay of 1 A of 30 GeV muons?

(a) The total power radiated is the energy per particle multiplied by the current divided by the charge per particle

$$\boxed{P = U_0 \frac{I}{e} = \frac{C_g E_0^4}{\rho} \frac{I}{e}}$$

(b) For electrons,

$$C_g = 8.85 \times 10^{-5} \text{ m} \cdot \text{GeV}^{-3}$$

Thus

$$P_e = \frac{\left(8.85 \times 10^{-5} \text{ m} \cdot \text{GeV}^{-3}\right)(30 \text{ GeV})^4}{250 \text{ m}} \frac{1 \text{ A}}{e} \Longrightarrow \boxed{P_e = 286.7 \text{ MW}}$$

For muons,

$$C_g = \frac{4\pi}{3}\frac{r_0}{(mc^2)^3}$$

To calculate the muon radius, we can take the electron radius and scale by the mass:

$$r_0 = \frac{q^2}{4\pi\epsilon_0 mc^2} = \frac{e^2}{4\pi\epsilon_0\,(105.6\text{ MeV})} = 1.3636\times10^{-17}\text{ m}$$

giving

$$C_g = \frac{4\pi}{3}\frac{1.3636\times10^{-17}\text{ m}}{(105.6\text{ MeV})^3} = 4.85\times10^{-14}\text{ m}\cdot\text{GeV}^{-3}$$

Thus

$$P_\mu = \frac{\left(4.85\times10^{-14}\text{ m}\cdot\text{GeV}^{-3}\right)(30\text{ GeV})^4}{250\text{ m}}\frac{1\text{ A}}{e} \implies \boxed{P_\mu = 0.1571\text{ W}}$$

(c) The muon decay lifetime in the lab frame is

$$\tau_{lab} = \tau_n\gamma = \left(2.1969811\times10^{-6}\text{ s}\right)\left(\frac{30\text{ GeV}}{105.6\text{ MeV}}\right) \implies \boxed{\tau_{lab} = 6.2414\times10^{-4}\text{ s}}$$

(d) In a muon decay, there is a continuum of possible particle kinetic energies because there are 3 products. Since we're concerned with a heat load, I'll consider the 'worst case scenario' where the electron carries away the full energy of the muon, $T_e = 30GeV$ (neglecting rest mass considerations). The power per decay is

$$P_\mu = \frac{30\text{ GeV}}{6.2414\ \times10^{-4}\text{ s}} = 7.701\times10^{-6}\text{ W}$$

The current is 1.0A, which corresponds to

$$\frac{N}{t} = \frac{1.0\text{ A}}{e} = 6.2415\times10^{18}\text{ s}^{-1}$$

If we take this current to be everywhere in the ring and use T,

$$N = \left(6.2415\times10^{18}\text{ s}^{-1}\right)\left(\frac{2\pi\,(250\text{ m})}{c}\right) = 3.2703\times10^{13}\text{ muons}$$

Thus the total power is

$$P = P_\mu N = \left(7.701\times10^{-6}\text{ W}\right)\left(3.2703\times10^{13}\right) \implies \boxed{P = 251.8\text{ MW}}$$

Alternatively, the energy stored in a beam with energy E_0, current I, and circulation time T is

$$U_{beam} = \frac{IT}{e}E_0$$

which gives $U_{beam} = 0.156$ MJ for the worst case scenario. The heat load is then

$$P_{heat} = \frac{U_{beam}}{\tau_\mu\gamma} = 250\text{ MW}$$

Problem 3. RF Phase Choice 15 pts.

In Lecture 11, we learned that for the continuous model with $qE_0 > 0$ a particle in a linear machine accelerates as:

$$q E_z(r=0, t=0) = qE_0 \cos(\varphi_s) > 0 \implies \text{accel}$$
$$q E_z(r=0, t=0) = qE_0 \cos(\varphi_s) < 0 \implies \text{deccel.}$$

with field that oscillates as $e^{i\omega t}$ and where from the Hamiltonian, we have the potential $V(\Delta\varphi)$ defined as

$$V(\Delta\varphi) = qE_0T[\sin(\varphi_s + \Delta\varphi) - \Delta\varphi\cos\varphi_s]; \quad qE_0T > 0.$$

Its concavity is given by

$$\left.\frac{\mathrm{d}^2V(\Delta\varphi)}{\mathrm{d}\varphi^2}\right|_{\varphi=\varphi_s} > 0 \implies \text{stability (longitudinal focusing)}$$
$$\left.\frac{\mathrm{d}^2V(\Delta\varphi)}{\mathrm{d}\varphi^2}\right|_{\varphi=\varphi_s} < 0 \implies \text{instability (longitudinal defocusing)}$$

locally about the synchronous particle. Figure 1 shows an example of $V(\Delta\varphi)$ and $\frac{\mathrm{d}^2V(\Delta\varphi)}{\mathrm{d}\varphi^2}$ at $\varphi_s = -30°$. Try plotting $V(\Delta\varphi)$ for different values of $\varphi_s = -30°$ for better visualization.

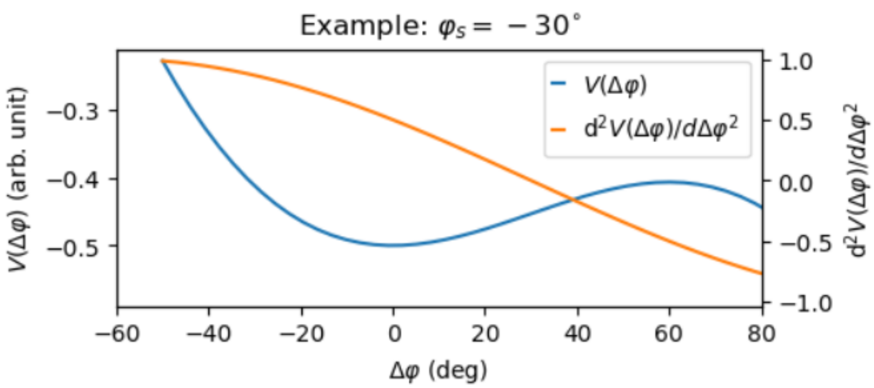

Figure 1: An example of $V(\Delta\varphi)$ and $\frac{\mathrm{d}^2V(\Delta\varphi)}{\mathrm{d}\varphi^2}$ at $\varphi_s = -30°$.

Use these to argue:

a) 5 pts: Range of φ_s for deceleration *and* focusing?

b) 5 pts: Range of φ_s for acceleration *and* defocusing?

c) 5 pts: Separately from parts a) and b), what value of φ_s will provide maximum longitudinal focusing and acceptance using the continuous model? Why? Does this allow acceleration?

SOLUTION: RF Phase Choice 10 pts.

Check Lecture 11 Sections 2.5.2 and 3.8 for a quick summary.

5 pts: For deceleration, $E_z < 0 \implies -\pi < \varphi_s < -\pi/2$.

Interval for decelerating and focusing: $-\pi < \varphi_s < -\pi/2$.

5 pts: For acceleration, $E_z > 0 \implies -\pi/2 < \varphi_s < \pi/2$.

Interval for accelerating and defocusing: $0 < \varphi_s < \pi/2$.

5 pts: $\varphi_s = -\pi/2$ or $\varphi_s = 3\pi/2$. Maximum positive slope (most positive $\mathrm{d}V/\mathrm{d}\varphi$). However, for $\varphi_s = -\pi/2$, $\cos(\varphi_s) = 0$ and there is no net acceleration.